

Finite Excess in Ordinal Nim Multiplication: Selected Order Reductions

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Abstract

Lenstra’s description of ordinal nim multiplication attaches to each odd prime p a finite excess m_p . Available values suggest that m_p is always 0, 1, or 4, with the value determined by the component support of $f(p) = \text{ord}_p(2)$. The universal rule remains open.

We reduce the rule to four exact multiplicative-order assertions. The zero arm asks a synchronized structural norm to generate every primary component of a primitive-support quotient. The ordinary singleton arm asks a marked cyclotomic unit to have full projective order at the unique prime over two selected by Conway ancestry. The cubic arm asks a recursively selected Gaussian period to generate a norm-one torus. The exceptional $2 \cdot 3^k$ arm asks an explicit corrected norm to have full primary order; equivalently, certain Capelli compositions are irreducible. The reductions retain the full primary valuation and the selected Frobenius phase.

For the cubic and exceptional arms we identify exact principal-ray and reduced-unit obstructions. For the singleton-even zero arm we identify a deterministic Conway–Fermat packet recursion and a supersingular realization; both expose the selected orbit without proving its order. We also obtain an unconditional level-dependent bound in the exceptional column. Generic trace, norm, conductor, class-number, factor-shape, or unselected reciprocity data do not evaluate the marked coordinate. Lean checks the algebraic reductions and named finite certificates, but defines rather than proves the universal exceptional target. Source-backed rows and exact finite calculations are stated separately from universal theorems.

1 Introduction

Conway’s nim addition and multiplication make the ordinals a proper class field of characteristic two. Its finite subfields are canonically embedded, and its first transfinite layers are assembled by explicit Kummer extensions [3, 7, 8]. Lenstra’s description associates to each odd prime p an element

$$\alpha_p = \kappa_{f(p)} + m_p, \quad f(p) = \text{ord}_p(2),$$

where $\kappa_{f(p)}$ is determined by the tower and $m_p \in \mathbb{N}$ is the *finite excess*. The same structure appears in the description of the ordinal fields On_p [5].

Let $\mathcal{Q}(h)$ be Lenstra’s component set for κ_h . The available data suggest the following rule.

Conjecture 1.1 (the 0/1/4 rule). *For every odd prime p ,*

$$m_p = \begin{cases} 0, & \mathcal{Q}(f(p)) \text{ is not a singleton odd prime power,} \\ 4, & f(p) = 2 \cdot 3^k \text{ for some } k \geq 1, \\ 1, & \text{otherwise.} \end{cases}$$

The term *selected* is essential. A finite field contains many elements having the same degree, trace, norm, or cyclotomic support, whereas the Conway construction recursively marks one Frobenius orbit. The open content is the primary multiplicative order of that orbit. Computations are

valuable for individual rows and as countermodels to overstrong lemmas, but they do not prove Conjecture 1.1.

After finite-field preliminaries, Section 3 gives the four-arm equivalence. Sections 4–7 develop one lossless selected coordinate for each arm. Section 9 separates the stronger 0/1/4 rule from absolute boundedness. Section 10 states the formal and finite-evidence boundary.

2 Finite-field preliminaries

For a finite algebraic number x , write

$$d(x) = [\mathbb{F}_2(x) : \mathbb{F}_2].$$

The element κ_h is the sum of the selected components κ_q , $q \in \mathcal{Q}(h)$, and $h \mid d(\kappa_h)$. The excess has the equivalent form

$$m_p = \min\{m \geq 0 : \kappa_{f(p)} + m \text{ is not a } p\text{-th power in its defining finite field}\}.$$

Finite integers are identified with their canonical finite numbers. We use standard facts about finite fields and Capelli composition [9].

Proposition 2.1 (exact power criterion). *Let p be prime, let $p \mid 2^E - 1$, and let $x \in \mathbb{F}_{2^E}^\times$. The following are equivalent:*

1. x is a p -th power;
2. $x^{(2^E-1)/p} = 1$;
3. $\text{ord}(x) \mid (2^E - 1)/p$.

Consequently, x is not a p -th power if and only if

$$v_p(\text{ord}(x)) = v_p(2^E - 1).$$

Proof. Choose a generator g of the cyclic multiplicative group and write $x = g^a$. Solvability of $py \equiv a \pmod{2^E - 1}$, the displayed power test, and the order divisibility are equivalent. The final statement is their p -primary form. $\square$

The valuation cannot be replaced by the weaker assertion $p \mid \text{ord}(x)$. If $h = f(p)$ and $h \mid E$, lifting the exponent gives

$$v_p(2^E - 1) = v_p(2^h - 1) + v_p(E/h).$$

Every order assertion below retains the full right-hand valuation.

Proposition 2.2 (norm reduction). *Let $h \mid E$, let $p \mid 2^h - 1$, and let $x \in \mathbb{F}_{2^E}^\times$. Then*

$$x^{(2^E-1)/p} = N_{\mathbb{F}_{2^E}/\mathbb{F}_{2^h}}(x)^{(2^h-1)/p}.$$

Thus x is a p -th power in $\mathbb{F}_{2^E}$ if and only if its norm is a p -th power in $\mathbb{F}_{2^h}$.

Proof. The norm exponent is $(2^E - 1)/(2^h - 1)$. Multiply it by $(2^h - 1)/p$ and apply Proposition 2.1. $\square$

3 The four exact arms

For $h > 1$, put

$$\Pi_h = \prod_{\substack{\ell \text{ odd prime} \\ \text{ord}_\ell(2)=h}} \ell^{v_\ell(2^h-1)}, \quad T_h = N_{\mathbb{F}_{2^{d(\kappa_h)}}/\mathbb{F}_{2^h}}(\kappa_h).$$

Thus Π_h is the full primary part supported at primes first occurring at residue degree h .

For an odd prime r , let $b_r = d(\alpha_r)$ and, for $a \geq 1$, put

$$x_{r,a} = \kappa_{r^a}, \quad F_{r,a} = \mathbb{F}_{2^{r^a b_r}}, \quad y_{r,a} = x_{r,a} + 1,$$

with $F_{r,0} = \mathbb{F}_{2^{b_r}}$. Define

$$G_{r,a} = F_{r,a}^\times / F_{r,a-1}^\times, \quad L_{r,a} = |G_{r,a}| = \Phi_{r^a}(2^{b_r}).$$

For $k \geq 1$, set $h = 3^k$, let $\zeta_k = \kappa_{3^k}$, and put

$$\gamma_k = \zeta_k + \zeta_k^{-1} \in \mathbb{F}_{2^h}^\times.$$

Finally, let $q = 2^h$, let $\omega = \zeta_k^h \in \mathbb{F}_4$, and define

$$A_k = \zeta_k^2 + \zeta_k + \omega, \quad M_k = A_k^{q-1}, \quad \Psi_k = \frac{\Phi_{2 \cdot 3^k}(2)}{3}.$$

Theorem 3.1 (four-arm reduction). *Conjecture 1.1 is equivalent to the conjunction of the following four universal assertions.*

Z. If $\mathcal{Q}(h)$ is not a singleton odd prime power, then $\Pi_h \mid \text{ord}(T_h)$.

O. If $r \neq 3$ is odd, $a \geq 1$, and $p \neq r$ is a prime divisor of $L_{r,a}$, then

$$v_p\left(\text{ord}(y_{r,a} F_{r,a-1}^\times)\right) = v_p(L_{r,a}).$$

C. The element γ_k is primitive in $\mathbb{F}_{2^{3^k}}^\times$ for every $k \geq 1$.

D. One has $\Psi_k \mid \text{ord}(M_k)$ for every $k \geq 1$.

Proof. Fix p and put $h = f(p)$. If $\mathcal{Q}(h)$ is not a singleton odd prime power, Proposition 4.1 below identifies $m_p = 0$ with the corresponding primary factor of Z .

If $\mathcal{Q}(h) = \{r^a\}$ with r odd, Lenstra's component formulas give $h = r^a s$ with $s \mid b_r$. Proposition 5.1 identifies the row primes with $p \neq r$ dividing $L_{r,a}$ and identifies $m_p = 1$ with assertion *O*.

When $r = 3$, one has $b_3 = 2$. The cases are $h = 3^a$ and $h = 2 \cdot 3^a$. Proposition 6.1 identifies the former with *C*, while Theorem 7.3 identifies the latter with *D*. These cases are disjoint and exhaustive. $\square$

Remark 3.2. Theorem 3.1 is an equivalence, not a proof of any of its four universal assertions. None of Z, O, C, D is presently known in full.

4 The zero arm

4.1 Synchronized structural norm

Proposition 4.1 (exact support criterion). *Let $E = d(\kappa_h)$. For an odd prime p with $f(p) = h$,*

$$m_p = 0 \iff v_p(\text{ord}(T_h)) = v_p(2^h - 1).$$

Consequently every such row has excess zero if and only if $\Pi_h \mid \text{ord}(T_h)$.

Proof. By the defining excess formula, $m_p = 0$ exactly when κ_h is not a p -th power. Proposition 2.2 moves the test to T_h , and Proposition 2.1 gives the primary valuation. $\square$

When κ_h has several components, T_h is not determined by separate component norms. Their common embedding carries a synchronized Frobenius phase. The following quotient removes every proper-subfield ambiguity while retaining that phase.

Proposition 4.2 (cubical primitive-support coordinate). *Let S be the set of prime divisors of h , let $M = \mathbb{F}_{2^h}$, and write $\varphi(x) = x^2$ and $\sigma_r = \varphi^{h/r}$. For $A \subseteq S$, put $\sigma_A = \prod_{r \in A} \sigma_r$, and define*

$$H_h = \prod_{r \in S} \mathbb{F}_{2^{h/r}}^\times, \quad \Delta_h(x) = \frac{\prod_{|A| \text{ even}} \sigma_A(x)}{\prod_{|A| \text{ odd}} \sigma_A(x)}.$$

Then

$$\ker \Delta_h = H_h, \quad |\text{im } \Delta_h| = \Phi_h(2),$$

and Δ_h induces an isomorphism $M^\times / H_h \simeq \text{im } \Delta_h$. Arm Z at degree h is equivalent to

$$\Pi_h \mid \text{ord}(\Delta_h(T_h)).$$

Proof. The operator is the power map

$$x \mapsto x^{\prod_{r|h, r \text{ prime}} (1 - 2^{h/r})}.$$

In the cyclic group of order $2^h - 1$, lifting the exponent prime by prime shows that its kernel has order $\text{lcm}_{r \in S} (2^{h/r} - 1)$, exactly the order of H_h . The cyclotomic product formula gives quotient order $\Phi_h(2)$. A prime with residue degree h divides no proper-subfield order, so the quotient preserves its full primary valuation. Proposition 4.1 gives the last assertion. $\square$

The cubical coordinate is canonical, but nonidentity is weaker than full primary order. Moreover, an additive cube made from independently shifted components is not the synchronized norm of their selected sum. Thus neither linear disjointness nor a nonzero raw cross-ratio proves Z .

4.2 The singleton-even Conway–Fermat chain

The zero arm also contains the singleton-even chain. Put

$$h_n = 2^{n+1}, \quad c_n = \kappa_{h_n} = 2^{2^n}, \quad E_n = \mathbb{F}_{2^{h_n}}, \quad F_n = 2^{2^n} + 1,$$

and $E_{-1} = \mathbb{F}_2$. The tower relation is

$$c_n^2 + c_n = \prod_{j < n} c_j.$$

Let

$$\delta_n = \text{ord}(c_n E_{n-1}^\times) \quad \text{in } E_n^\times / E_{n-1}^\times.$$

Theorem 4.3 (Conway–Fermat selected order). *One has $1 < \delta_n \mid F_n$, and the zero arm at every prime divisor of F_n is equivalent to*

$$\delta_n = F_n \quad (n \geq 0).$$

All compatible binary root choices form one absolute-Frobenius orbit, so they have the same order vector $(\delta_0, \dots, \delta_n)$. The remaining assertion is therefore intrinsic to the selected tower rather than to an extra choice of root.

Proof. The quotient group has order $(2^{2^{n+1}} - 1)/(2^{2^n} - 1) = F_n$, and $c_n \notin E_{n-1}$. Every prime $p \mid F_n$ has $f(p) = 2^{n+1}$, with its full primary part in this quotient. Proposition 2.1 gives the displayed equality for δ_n .

A compatible prefix has two extensions at each level, so there are 2^{n+1} compatible tuples. Frobenius preserves the defining equations, and the 2^{n+1} Frobenius conjugates are distinct because c_n has that degree. They exhaust the tuples and preserve all relative orders. $\square$

The recurrence has a deterministic packet form. If $u_n = (c_n + 1)/c_n$, then $\text{ord}(u_n) = \delta_n$ and $a_{n-1} = \prod_{j < n} c_j = u_n/(u_n + 1)^2$. For each divisor $e > 1$ of F_n , the inverse pairs of primitive e -th roots give a squarefree packet polynomial

$$C_e(X) = \prod_{[z] \in \mu_e^{\text{prim}}/(z \sim z^{-1})} \left(X - \frac{z}{(z+1)^2} \right) \in \mathbb{F}_2[X].$$

Every irreducible factor has degree 2^n , and

$$\delta_n = e \iff \text{the selected minimal polynomial of } a_{n-1} \text{ divides } C_e.$$

If P is such a factor, its unique irreducible child is

$$\text{Res}_X(P(X), Y^2 + XY + X^3).$$

Thus the literal chain is one deterministic orbit beginning at $X + 1$. Full parent conductor does not imply full child conductor: exact degree-16 packet examples have identical full parent order but deterministic children of different conductor. Consequently any induction must use the complete marked orbit, not only the integer conductor.

This isolates the part not settled by the existing Conway-tower literature. Popovych proves lower bounds and verifies the required relative full orders through level eleven using the known Fermat factorizations [11]. Cagliero, Herman, and Szechtman prove an exact product formula for global order in terms of the relative orders, as well as a normal basis theorem for the associated tower elements [2]. Neither result forces every relative factor δ_n to equal F_n ; that all-level equality is precisely the remaining selected assertion here.

There is an exact supersingular realization of the selected ancestry.

Proposition 4.4 (supersingular realization and pairing boundary). *On*

$$\mathcal{E} : \quad y^2 + y = x^3 + x^2$$

put $P_n = (c_n, c_{n+1})$. Then $P_n \in \mathcal{E}$, and for $n \geq 2$,

$$[F_n]P_n = O.$$

More precisely, if π is absolute Frobenius and $\iota = \pi + [1]$, then

$$\iota(x, y) = (x + 1, y + x + 1), \quad \iota^2 = [-1], \quad \pi^4 = [-4].$$

For $g = (y + 1)/y$, one has

$$g(P_n) = \frac{c_{n+1} + 1}{c_{n+1}}, \quad \text{ord}(g(P_n)) = \delta_{n+1}.$$

If $T = (0, 0)$, then T has order five and

$$\text{div}(g) = 2[-T] + [2T] - 2[T] - [-2T].$$

Thus g is the quotient of the two fixed doubling Miller functions at $-T$ and T , but it is not a Weil or reduced-Tate pairing function attached to the F_n -torsion point P_n when $n \geq 2$.

Proof. The Conway recursion gives

$$c_{n+1}^2 + c_{n+1} = c_n(c_n^2 + c_n) = c_n^3 + c_n^2,$$

so P_n lies on $\mathcal{E}$. The coordinate formula for $\iota(P) = \pi(P) + P$ follows from the chord law. Applying it twice gives the inverse $(x, y) \mapsto (x, y + 1)$, so $\pi^2 + 2\pi + 2 = 0$ in $\text{End}(\mathcal{E})$ and hence $\pi^4 = [-4]$.

Put $H = 2^{n+1}$. The 2^H -Frobenius fixes c_n and sends c_{n+1} to $c_{n+1} + 1$, hence sends P_n to $-P_n$. For $n \geq 2$,

$$\pi^H = (\pi^4)^{2^{n-1}} = [2^{2^n}],$$

which gives $[2^{2^n} + 1]P_n = O$.

The formula for $g(P_n)$ is immediate from the coordinates, and its order is δ_{n+1} by the definition of u_{n+1} . The five rational points of $\mathcal{E}/\mathbb{F}_2$ show that T has order five. Direct intersection with the two horizontal lines gives

$$\text{div}(y) = 2[T] + [-2T] - 3[O], \quad \text{div}(y + 1) = 2[-T] + [2T] - 3[O],$$

and subtraction gives the divisor of g . Equivalently, $g = f_{2,-T}/f_{2,T}$ for functions normalized at O . Its divisor is supported on fixed rational five-torsion, with coefficients $2, 1, -2, -1$. For $n \geq 2$, these valuations are not divisible by $F_n \geq 17$, whereas the divisor of an F_n -Miller function and every F_n -th-power correction have all coefficients divisible by F_n . Thus g cannot be such a function attached to P_n , even modulo an F_n -th power. A Weil or reduced Tate pairing on $\mathcal{E}[F_n]$ takes values in μ_{F_n} , whereas $g(P_n)$ has order dividing F_{n+1} . Pairwise coprimality of the Fermat numbers and $\delta_{n+1} > 1$ exclude equality. $\square$

The proposition supplies exact CM annihilator control but not generator status. For $n \geq 3$, no elliptic endomorphism can carry P_n to P_{n-1} : the image order would divide both coprime numbers F_n and F_{n-1} , although $P_{n-1} \neq O$. For $n = 2$, the same conclusion follows from $\text{ord}(P_1) = 3$ and $\text{ord}(P_2) \mid 17$. Thus the relation $x(P_n) = y(P_{n-1})$ is not an isogeny division tower, and pairing functoriality along this proposed tower cannot propagate primitivity. A perfect alternating pairing likewise guarantees some detecting partner, not nonvanishing on a prescribed line.

The tempting cubic curve $Y^2 + Y = X^3$ is not a coordinate rewrite of $\mathcal{E}$: the two curves have respectively three and five rational points over $\mathbb{F}_2$. For $u = (Y + 1)/Y$ on that auxiliary curve, $u/(u + 1)^2 = X^3$; every positive-level Fermat-torsion value is already a cube because $\gcd(F_n, 3) = 1$. Its natural cubic Kummer class is therefore trivial and cannot replace the selected five-torsion function above. The remaining statement is still exclusion of every proper-conductor packet along the complete resultant chain. Trace, relative norm, irreducibility, and the etale Jacobian hold for proper packets as well as the full packet.

5 The ordinary singleton arm

Fix an odd prime r , write $b = b_r$, and abbreviate $x_a = x_{r,a}$, $y_a = x_a + 1$, and $F_a = F_{r,a}$.

Proposition 5.1 (projective order criterion). *For $a \geq 1$,*

$$N_{F_a/F_{a-1}}(y_a) = y_{a-1}, \quad |F_a^\times / F_{a-1}^\times| = L_{r,a}.$$

If $p \neq r$ divides $L_{r,a}$, then the corresponding singleton row has

$$m_p = 1 \iff v_p(\text{ord}(y_a F_{a-1}^\times)) = v_p(L_{r,a}).$$

Every singleton row with component r^a arises this way.

Proof. The relative conjugates of x_a are ξx_a for $\xi \in \mu_r$; evaluating their binomial at one gives the norm identity. The quotient order is the quotient of the two finite-field unit orders. The cyclotomic divisor criterion gives $\text{ord}_p(2^b) = r^a$ for $p \neq r$ dividing $L_{r,a}$, and Lenstra's component theorem identifies precisely the singleton rows. The lower unit group has no p -part, so Proposition 2.1 gives the last equivalence. $\square$

Let $N_a = \text{ord}(x_a)$, choose a primitive complex N_a -th root ξ_{N_a} , and let $\mathfrak{P}_a \mid 2$ be the prime whose residue sends ξ_{N_a} to x_a . Put $\varepsilon_a = 1 - \xi_{N_a}$.

Theorem 5.2 (selected cyclotomic-unit form). *For $p \neq r$ dividing $L_{r,a}$, arm O at (r, a, p) is equivalent to*

$$\left(\frac{\varepsilon_a}{\mathfrak{P}_a} \right)_p \neq 1.$$

The compatible primes $\mathfrak{P}_a$ form a unique inert ray above two: $\mathfrak{P}_a$ is the unique prime above $\mathfrak{P}_{a-1}$, with relative residue degree r , and

$$N_{\mathbb{Q}(\xi_{N_a})/\mathbb{Q}(\xi_{N_{a-1}})}(1 - \xi_{N_a}) = 1 - \xi_{N_{a-1}}.$$

Nevertheless, residue-field corestriction kills the new p -coordinate and does not determine the displayed residue symbol.

Proof. The reduction of ε_a is y_a . The unramified power-residue formula gives

$$\left(\frac{\varepsilon_a}{\mathfrak{P}_a} \right)_p = y_a^{(|F_a|-1)/p},$$

so Proposition 2.1 and Proposition 5.1 give the equivalence.

The Kummer steps increase the r -primary part of N_a by one and give $\text{ord}_{N_a}(2) = br^a$. The relative cyclotomic degree and the residue-degree ratio are therefore both r , forcing one prime above the preceding selected prime. The global norm identity is the product over the r -th roots of unity. Finally, $p \mid \Phi_{r^a}(2^b)$ but $p \nmid 2^{br^{a-1}} - 1$, so the lower residue field has trivial p -Kummer quotient. Its norm receives no information about the new selected class. $\square$

The selected symbol has an exact Jacobi–Fourier expansion. Let $Q = |F_a|$, let $H = \langle x_a \rangle \subset F_a^\times$ have order N_a , put $M = (Q - 1)/N_a$, and let χ be a complex-valued order- p power character. Extend complex-valued multiplicative characters by zero at zero and define

$$f(h) = \chi(1 - h), \quad \hat{f}(\theta) = \sum_{h \in H} f(h) \theta(h)^{-1} \quad (\theta \in \hat{H}).$$

For an extension ψ_θ of θ to $F_a^\times$, and with $H^\perp$ the character group of $F_a^\times/H$, one has

$$\widehat{f}(\theta) = \frac{1}{M} \sum_{\lambda \in H^\perp} J(\psi_\theta^{-1} \lambda, \chi), \quad f(x_a) = \frac{1}{N_a} \sum_{\theta \in \widehat{H}} \theta(x_a) \widehat{f}(\theta).$$

Indeed, the character average $M^{-1} \sum_{\lambda \in H^\perp} \lambda$ is the indicator of H ; inserting it in the first Fourier sum gives the Jacobi sums, and Fourier inversion gives the second identity. This is a fully explicit reciprocity formula, but the selected label survives in $\theta(x_a)$ and the Jacobi unit phases. Ideal factorization or valuation formulas for the Jacobi sums do not by themselves determine this inverse-DFT entry. This separation of the ideal and p -adic unit conditions is also visible in Uehara's cyclotomic-unit criterion [14]; it supplies an exact test, not a uniform nonvanishing theorem for the Conway-selected entry.

This residue-symbol condition is a marked Artin-symbol assertion, not a statement about decomposition or conductor. Its ray character splits into a ramified local logarithm and an unramified ideal-class character evaluated at $[\mathfrak{P}_a]$. Stickelberger annihilation reduces the latter to an explicit decomposition-invariant quotient, but does not make it vanish uniformly. This is a genuine boundary: at the actual Conway conductor $r = 11$, $a = 1$, $p = 23$, one contributing Stickelberger scalar vanishes with exact 23-adic valuation one. The corresponding unramified degree-23 class line is nontrivial, and every prime over two has nontrivial Artin symbol on it, although the literal row still has $m_{23} = 1$. Thus successful rows need not have a zero unramified term; the open assertion is noncancellation in the actual selected sum.

6 The cubic arm

Let $h = 3^k$, $\zeta_k = \kappa_{3^k}$, and $\gamma_k = \zeta_k + \zeta_k^{-1}$.

Proposition 6.1 (half-angle reduction). *The element ζ_k is a primitive 3^{k+1} -st root of unity,*

$$\mathbb{F}_2(\zeta_k) = \mathbb{F}_{2^{2h}}, \quad \mathbb{F}_2(\gamma_k) = \mathbb{F}_{2^h}.$$

Moreover,

$$\text{ord}(\zeta_k + 1) = 3^{k+1} \text{ord}(\gamma_k),$$

and the translates by 2 and 3 have the same order. Hence the rows with $f(p) = 3^k$ have excess one precisely when γ_k has the full relevant primary order; over all levels this is equivalent to primitivity of every γ_k .

Proof. The defining Kummer equation gives $\zeta_k^{3^k} = \kappa_2$, a primitive cube root. The order of two modulo 3^{k+1} is $2h$, and the least exponent giving ± 1 is h , proving the field degrees. If s is the inverse of two modulo 3^{k+1} , then

$$\zeta_k + 1 = \zeta_k^s (\zeta_k^s + \zeta_k^{-s}), \quad (\zeta_k^s + \zeta_k^{-s})^2 = \gamma_k.$$

The factors lie in the coprime circle and real parts of the unit group, giving the order identity. Frobenius gives the other translates, and Proposition 2.1 finishes the reduction. $\square$

Put $q_k = 2^{3^{k-1}}$ and

$$U_k = \ker \left(N_{\mathbb{F}_{q_k^3}/\mathbb{F}_{q_k}} \right), \quad \eta_k = \gamma_k^{q_k-1}, \quad \eta_0 = 0.$$

Theorem 6.2 (selected norm-one coordinate). *One has $|U_k| = q_k^2 + q_k + 1 = \Phi_{3^k}(2)$. The element η_k generates $\mathbb{F}_{2^{3^k}}$ and is the selected root of*

$$X^3 + \eta_{k-1}X^2 + (\eta_{k-1} + 1)X + 1.$$

Furthermore,

$$C_k \iff C_{k-1} \text{ and } \text{ord}(\eta_k) = |U_k|,$$

where C_j denotes primitivity of γ_j .

Proof. The definition puts η_k in the norm-one group. Frobenius calculation from ζ_k gives the displayed cubic; its coefficients are respectively the trace, second symmetric function, and norm. A base-field element of U_k has order dividing $\gcd(q_k - 1, 3) = 1$, so the nontrivial cubic is irreducible. Its quadratic coefficient recursively recovers η_{k-1} , proving the field degree. Finally, the finite-field unit group splits into its base-field and norm-one factors, whose orders are coprime. The norm propagates the old part of γ_k and η_k detects exactly the new factor. $\square$

The recursion selects one Frobenius orbit inside a Singer difference set. Indeed, for

$$R(x) = \frac{x^3 + x + 1}{x^2 + x}, \quad W(x) = \frac{x + \omega}{x + \omega^2},$$

one has

$$R(\eta_k) = \eta_{k-1}, \quad W(R(x)) = W(x)^3, \quad W(\eta_k) = \zeta_k^2.$$

Thus the complete lower ancestry, not merely degree or trace, marks the orbit. The displayed conjugacy exposes its cyclotomic phase but does not evaluate the multiplicative order of η_k .

Proposition 6.3 (selected Singer sieve). *Let $N = |U_k|$ and $h = 3^k$. Then*

$$\text{ord}(\eta_k) \geq h(h - 1) + 1.$$

If a prime $\ell \mid N$ is missing from the selected order at its full valuation, then

$$\frac{N}{\ell} \geq h(h - 1) + 1.$$

Hence every prime $\ell > N/(h(h - 1) + 1)$ occurs with full valuation.

Proof. The selected Frobenius orbit has h points in the Singer $(N, q_k + 1, 1)$ difference set. Its $h(h - 1)$ ordered off-diagonal quotients are distinct nonidentity elements of $\langle \eta_k \rangle$, giving the first bound. If the full ℓ -part is missing, the orbit lies in the subgroup of order N/ℓ , which must contain all these quotients. $\square$

The sieve is polynomial in h , while N is exponential. It proves a genuine selected large-factor range but not arm C .

There is an exact number-field factorization of the remaining index. Put $L = \mathbb{Q}(\zeta_{3^{k+1}})^+$ and $K = \mathbb{Q}(\zeta_{3^k})^+$, reduce global units E_F and circular units C_F modulo the inert prime over two, and write

$$R_F = (\mathcal{O}_F/2)^\times / \overline{E}_F, \quad T_F = \overline{E}_F / \overline{C}_F.$$

Then

$$[U_k : \langle \eta_k \rangle] = |\ker(N : R_L \rightarrow R_K)| \frac{[\overline{E}_L : \overline{C}_L]}{[\overline{E}_K : \overline{C}_K]}.$$

The first factor is a relative principal 2-ray kernel; the second divides the relative real class-number quotient. Even class number one would leave the principal-ray factor. This is not a formal possibility only: analogous prime-conductor periods can have inert two and class number one while their finite circular-unit reduction has nontrivial defect [6]. Arm C is exactly the assertion that both factors in the displayed index formula vanish at every level.

7 The exceptional arm

Let $h = 3^k$, $q = 2^h$, $\zeta = \zeta_k$, and $\omega = \zeta^h \in \mathbb{F}_4$.

Proposition 7.1 (first possible translate). *If $f(p) = 2 \cdot 3^k$, then $m_p \geq 4$.*

Proof. Such a prime divides $2^h + 1$ but neither 3^{k+1} nor $2^h - 1$. The half-angle decomposition in Proposition 6.1 shows that each of $\zeta, \zeta + 1, \zeta + 2, \zeta + 3$ has order prime to p . All four are p -th powers in their defining fields, so the excess definition gives the claim. $\square$

The translate by four lies in a compositum with $\mathbb{F}_{16}$. The nontrivial relative Frobenius over $\mathbb{F}_4$ exchanges the numbers 4 and 5, which are the roots of $Y^2 + Y + \omega$.

Proposition 7.2 (corrected norm). *In $\mathbb{F}_{2^{4h}}/\mathbb{F}_{2^{2h}}$,*

$$N(\zeta + 4) = (\zeta + 4)(\zeta + 5) = \zeta^2 + \zeta + \omega = A_k.$$

Proof. One has $4 + 5 = 1$ and $4 \cdot 5 = \omega$. Expanding the product gives the formula. $\square$

Define

$$R_k(Y) = \text{Res}_X \left(X^{2h} + X^h + 1, Y + X^h + X^2 + X \right).$$

Theorem 7.3 (exact exceptional target). *The polynomial R_k is the irreducible degree- $2h$ minimal polynomial of A_k . The prime divisors of $\Psi_k = \Phi_{2 \cdot 3^k}(2)/3$ are exactly the primes $\ell \neq 3$ with $\text{ord}_\ell(2) = 2h$, with their full valuations. The following are equivalent:*

1. *every row with $f(\ell) = 2 \cdot 3^k$ has $m_\ell = 4$;*
2. *$\Psi_k \mid \text{ord}(M_k)$;*
3. *A_k is not an ℓ -th power in $\mathbb{F}_{2^{2h}}$ for every prime $\ell \mid \Psi_k$;*
4. *$R_k(X^\ell)$ is irreducible over $\mathbb{F}_2$ for every prime $\ell \mid \Psi_k$.*

Proof. The element ζ has minimal polynomial $X^{2h} + X^h + 1$. Comparing the support of a hypothetical Frobenius relation $A_k^{2^j} = A_k$ modulo this trinomial forces $2^j \equiv 1 \pmod{3h}$, so A_k has degree $2h$ and the resultant is its minimal polynomial.

The cyclotomic divisor criterion applied to $\Phi_{2 \cdot 3^k}(2)$ gives the stated prime support after removing its single factor of three. Proposition 7.1 and the corrected norm give

$$m_\ell = 4 \iff A_k \notin (\mathbb{F}_{2^{2h}}^\times)^\ell.$$

Raising to $q - 1$ is invertible on the current ℓ -Kummer quotient, so this is equivalent to the full ℓ -primary part occurring in $\text{ord}(M_k)$. Finally, μ_ℓ lies in the defining field; Kummer theory and Capelli's lemma identify nonpower with irreducibility of $R_k(X^\ell)$. $\square$

Proposition 7.4 (effective exceptional bound). *Let $k \geq 2$, $h = 3^k$, and let $\ell \mid \Psi_k$ be prime. If t is the least power of two satisfying $t \geq \lceil \log_2(h-1) \rceil$, then*

$$m_\ell \leq 4^t - 1 < 16h^4.$$

Proof. Put $K = \mathbb{F}_{4^h}$ and extend constants to $L = K\mathbb{F}_{4^t}$. For a character χ of exact order ℓ , the sum

$$S = \sum_{c \in \mathbb{F}_{4^t}} (\chi \circ N_{L/K})(\zeta + c)$$

equals 4^t precisely when every translate is an ℓ -th power. The associated primitive even function-field character has L -polynomial $(1 - T) \prod_{j=1}^{h-2} (1 - \alpha_j T)$ with $|\alpha_j| = 2$. Constant-field extension therefore gives

$$|S| \leq 1 + (h-2)2^t < 4^t$$

when $2^t \geq h-1$; see [12]. Some translate is a nonpower, and norm injectivity on the ℓ -Kummer quotient returns the statement to its defining field. The elements of $\mathbb{F}_{4^t}$ are the finite numbers below 4^t . The displayed polynomial estimate follows from the minimal choice of t . $\square$

This bound is level-dependent and does not prove absolute boundedness or the value four.

The exceptional coordinate also has an exact circular-ray factorization. In the real field generated by roots of orders 3^{k+1} and 5, let $\mathfrak{P}$ be the prime selected by $\zeta \mapsto \zeta_k$ and the conductor-five coordinate $\mapsto \omega$. Let E_L and C_L be the global and circular units and let a subscript ℓ denote the primary projection after reduction. Then

$$\overline{C}_{L,\ell} = \langle A_{k,\ell} \rangle = \langle M_{k,\ell} \rangle,$$

and

$$[k_{\mathfrak{P},\ell} : \langle M_{k,\ell} \rangle] = |\ker(\text{Cl}_L(\mathfrak{P})_\ell \rightarrow \text{Cl}_L)_\ell| \cdot |(\overline{E}_L / \overline{C}_L)_\ell|.$$

The second factor is controlled by the real class number, but the first is a principal ray kernel. The marked conductor-five cyclotomic unit is globally not an ℓ -th power; the height theorem of Amoroso–Dvornicich gives this directly [1]. Global nonpower does not determine its Frobenius at the selected prime. Arm D is exactly the assertion that both factors in the displayed ray-class index vanish.

The current C and D coordinates are orthogonal. If $q_0 = 2^{h/3}$ and $q = q_0^3$, their norm-one tori have orders

$$q_0^2 + q_0 + 1 \quad \text{and} \quad q + 1,$$

which are coprime. The exceptional cubical operator kills the cubic subfield, while the local norm kills the exceptional torus. Hence the two principal-ray factors cannot force one another through the shared prime over two. Each requires its own marked unit-image calculation.

Even a perfect one-dimensional Kummer–Artin pairing does not show that a prescribed prime has nonzero image in the Artin quotient. The obstruction already occurs arithmetically.

Proposition 7.5 (perfect-pairing no-go). *There exist a number field K containing μ_3 , a global unit u which is not a cube in K , and exactly one prime $\mathfrak{q}$ of K above two; this prime splits completely in the nontrivial cyclic Kummer extension $K(\sqrt[3]{u})/K$.*

Proof. Take

$$K = \mathbb{Q}(\zeta_{12}) = \mathbb{Q}(\zeta_3, \sqrt{3}), \quad u = 2 + \sqrt{3}.$$

The Pell equation gives $\mathcal{O}_{\mathbb{Q}(\sqrt{3})}^\times = \{\pm u^n : n \in \mathbb{Z}\}$, so u is not a cube in the real quadratic subfield. If $X^3 - u$ acquired a root in its quadratic extension K , its cubic factorization over the real subfield would also have a linear factor. Hence u is not a cube in K , and adjoining a cube root gives a nontrivial cyclic cubic extension.

Locally at two, $\mathbb{Q}_2(\zeta_3)/\mathbb{Q}_2$ is unramified quadratic and adjoining i is ramified quadratic. The two local quadratics are linearly disjoint, so $K \otimes \mathbb{Q}_2$ is a degree-four field; hence two has a unique prime $\mathfrak{q}$ in K . In its residue field $\sqrt{3} \equiv 1$, so $u \equiv 1 \pmod{\mathfrak{q}}$. The polynomial $X^3 - u$ has the simple root 1 modulo $\mathfrak{q}$, because its derivative at one is the unit 3. Hensel's lemma makes u a cube in $K_{\mathfrak{q}}$; since $\mu_3 \subset K_{\mathfrak{q}}$, the polynomial splits completely there and $\mathfrak{q}$ splits completely in $K(\sqrt[3]{u})$. $\square$

Thus nontriviality of the Kummer line, one-dimensionality of its dual, and even uniqueness of the prime above two are compatible with zero marked pairing. In the four arms, perfect duality reduces the target to nonvanishing of the Conway-selected prime class; it does not prove that nonvanishing.

8 Nim reciprocity saturation and marked holonomy

The residue-symbol reductions suggest using reciprocity to evaluate the remaining phase. There is a precise saturation boundary for that program. Let ℓ be an odd prime, let $s \in \mu_\ell$, and put $E = \text{ord}_\ell(2)$. The two-axis Frobenius–cyclotomic family is $s_{j,c} = s^{2^j c}$ for $0 \leq j < E$ and $c \in \mathbb{F}_\ell^\times$.

Proposition 8.1 (reciprocity saturation). *The following statements hold.*

1. Every multiplicative word in the $s_{j,c}$ is s^λ , where

$$\lambda = \sum_{j,c} w_{j,c} 2^j c \pmod{\ell}.$$

It is blind when $\lambda = 0$ and, when $\lambda \neq 0$, equals one exactly when the original marked phase equals one.

2. For multiplicative characters χ, ψ of a finite field and $a \neq 0$, the Jacobi convolution

$$J_{\chi,\psi}(a) = \sum_x \chi(x) \psi(a - x)$$

satisfies

$$J_{\chi,\psi}(a) = \chi(a) \psi(a) J_{\chi,\psi}(1).$$

In particular, if both characters are powers of one order- ℓ character, the marked dependence is again one power of the same phase.

3. In the affine group on $\mathbb{F}_\ell$, write (t, q) , with $q \in \mathbb{F}_\ell^\times$, for the map $x \mapsto qx + t$. If $q \neq 1$, all (t, q) are conjugate. Consequently every class function is independent of t .

Proof. The first statement follows by collecting exponents. For the second, substitute $x = ay$ and factor $\chi(a)\psi(a)$ from every summand. For the third, conjugation by the translation $x \mapsto x + a$ changes t to $t + (1 - q)a$; taking $a = t/(q - 1)$ makes this zero. $\square$

Thus norm products, weighted Stickelberger words, and Gauss or Jacobi transforms either erase or re-encode one phase. A class function on the single fixed-multiplier affine coset is blinder still: it forgets the translation coordinate in part 3. None of these operations independently evaluates the selected phase.

Ordinary cross-arm multiplicative words face the same boundary. If a strict ancestor lies in a field $\mathbb{F}_{2^d}$ with $f(\ell) \nmid d$, exactness of the residue degree gives $\ell \nmid 2^d - 1$; the ℓ -power map is therefore an automorphism on that ancestor group and its ℓ -Kummer image is trivial. Thus every cross-arm word built from strict lower-level coordinates in such degrees separates its coprime-primary torsion. Nor does the next ℓ -component give an induction: its defining radicand is $\alpha_\ell = \kappa_{f(\ell)} + m_\ell$, so irreducibility proves only that the actual winning translate is a non- ℓ -th power. Using that fact to identify m_ℓ with the conjectural 0, 1, or 4 would already assume the target.

The intrinsic coordinate is a full Frobenius-orbit holonomy, not a one-step affine translation. Let $\beta \in \mathbb{F}_{2^E}^\times$ be the selected element, put $s = \beta^{(2^E-1)/\ell}$, choose $\alpha^\ell = \beta$ in an algebraic closure, fix a primitive $\zeta \in \mu_\ell$, and let $\varphi(z) = z^2$. Then

$$\frac{\varphi^E(\alpha)}{\alpha} = \alpha^{2^E-1} = \beta^{(2^E-1)/\ell} = s.$$

This ratio is unchanged when α is replaced by $\zeta^a \alpha$, because φ^E fixes μ_ℓ . If $s = \zeta^T$, the E -step action on the labeled roots is the pure translation $e_x \mapsto e_{x+T}$. Hence on the Fourier lines

$$v_r = \sum_x \zeta^{-rx} e_x, \quad \varphi^E(v_r) = \zeta^{rT} v_r.$$

The diagonal eigenvalue at $r = 1$ is exactly the selected Kummer–Artin symbol.

Equivalently, choose a root frame over each conjugate $\beta_j = \beta^{2^j}$ and write the one-step maps as $e_{j,x} \mapsto e_{j+1, 2x+t_j}$. Their cyclic holonomy is

$$T = \sum_{j=0}^{E-1} 2^{E-1-j} t_j \pmod{\ell}.$$

Changing frames by $\alpha_j \mapsto \zeta^{a_j} \alpha_j$ sends t_j to $t_j + 2a_j - a_{j+1}$; with $a_E = a_0$, these changes telescope because $2^E = 1$ modulo ℓ . Thus T is intrinsic, while the individual off-diagonal coefficients

$$\rho(t_j, 2)v_r = \zeta^{r2^{-1}t_j} v_{r2^{-1}}$$

are gauge-dependent. Every proper edge can be normalized away recursively, but a closed gauge cannot erase the cyclic holonomy: all but one edge may be made trivial, with the remaining edge carrying T .

This also explains the cocycle no-go. A homomorphic splitting on a commutative multiplier subgroup obeys $t(qr) = t(q) + qt(r)$. Fixing $q_0 \neq 1$ and comparing $t(qq_0)$ with $t(q_0q)$ gives

$$t(q) = (1 - q)a, \quad a = \frac{t(q_0)}{1 - q_0}.$$

Every such cocycle is a coboundary and has trivial full-period transfer. Powers of one affine lift give this law on the subgroup generated by its multiplier. A general labeled section has a factor set, but changing root frames changes that factor set by a coboundary; only its cyclic holonomy is gauge-invariant.

Affine ancestry between two such torsor paths cannot create a missing phase. If the vertex maps are $x \mapsto \lambda x + b_j$, then the target translations have the form

$$u_j = \lambda t_j + b_{j+1} - 2b_j,$$

and the path holonomies satisfy

$$U_n = \lambda T_n + b_n - 2^n b_0.$$

On a closed E -cycle the endpoint term telescopes, so $U = \lambda T$. Consequently an affine recursion from a strict ancestor with trivial ℓ -holonomy cannot prove a nonzero current phase; a successful relation must contain a genuinely non-affine or independently nonzero closing source.

A successful nim-reciprocity proof must therefore retain and independently evaluate the marked holonomy T , equivalently s , along the literal Conway ancestry. Constructing the holonomy from the selected phase is an exact reformulation, not yet its evaluation.

9 Boundedness and the remaining theorem

Lenstra asked whether the finite excesses are absolutely bounded [7]. Conjecture 1.1 would give the bound four, but boundedness is strictly weaker than the maximal-order assertions above.

For $a \geq 0$, let $C_a = \mathbb{F}_{2^{2^a}}$. For an odd prime p , put $x_p = \kappa_{f(p)}$ and $L_{p,a} = \mathbb{F}_2(x_p)C_a$.

Proposition 9.1 (finite-line criterion). *For every p and a ,*

$$m_p < 2^{2^a} \iff \text{some } c \in C_a \text{ has } x_p + c \notin L_{p,a}^p.$$

Consequently,

$$\sup_p m_p < \infty \iff \text{one fixed } a \text{ works for every } p.$$

Proof. For $F_c = \mathbb{F}_2(x_p + c)$, one has $F_c C_a = L_{p,a}$ and $[L_{p,a} : F_c] \mid 2^a$, which is prime to p . Extension is therefore injective on p -Kummer quotients. The finite numbers below 2^{2^a} are exactly C_a , proving the first equivalence from the excess definition. A fixed a gives a uniform bound; conversely choose a above any given uniform bound. $\square$

No fixed finite affine line is known to work for every prime. The complete universal frontier remains the four selected order assertions in Theorem 3.1:

1. full primary order for the synchronized structural norm, including every Conway–Fermat equality $\delta_n = F_n$;
2. full primary projective order for the marked ordinary cyclotomic unit;
3. primitivity of every selected Gaussian period γ_k , equivalently generation by every η_k ;
4. $\Psi_k \mid \text{ord}(M_k)$ at every exceptional level.

Each is now one-dimensional and ancestry-sensitive. Degree, trace, norm, conductor, factor parity, or a statement about generic ambient points does not decide it.

Formal group	Kernel-checked content
Kummer interface	First-nonpower characterization, cyclic-group and finite-field power tests with full primary valuation, quotient order, and norm transport.
Four-arm algebra	Corrected exceptional norm, toric and Dickson semiconjugacies, selected cubic recurrence, cubical Z/D identities, and C/D orthogonality.
Reciprocity saturation	Two-axis Frobenius-word reduction, oriented-prefix cocycle, affine class-function blindness, marked Jacobi scaling, labeled Fourier-line pullback, and gauge-invariant path holonomy.
Singleton-even boundary	Packet substitutions and norms, child-image equation, ancestry derivative, Dickson–Lucas tangency factorization, and the distinction between Conway and Wiedemann Laurent recursions, together with the auxiliary cubic-class no-go, abstract prescribed-pairing countermodel, and coprime-annihilator map obstruction.
Finite certificates	Recorded primality, factor, order, modular Stickelberger, and selected binary Artin-phase calculations supplied to the formal checker.

Table 1: Lean-checked content relevant to the excess reduction.

10 Formal verification and finite evidence

The formal development uses Lean 4 [4] and Mathlib [13].

Its import-only entry point is `Ogdoad.Papers.Excess`. It checks the algebraic content summarized in Table 1.

The identifier `DPrimeTarget` defines the universal assertion D . It is neither assumed nor proved. The formalization proves ingredients of Theorem 3.1; it does not contain an end-to-end proof of Conjecture 1.1. Supplied finite coefficients and factorizations are checked as data, not derived from their external number-theoretic sources.

The finite evidence has the following status.

- The 126 values for odd primes $3 \leq p \leq 709$ are source-backed by OEIS A380496 [10]. They are tabulated data, not a universal theorem.
- The rows $m_{359} = m_{719} = m_{727} = 1$ have separate exact local finite-field certificates.
- In the cubic arm, C_1, C_2, C_3 have exact order proofs and C_4, C_5 have complete finite certificates. C_6 retains a source-assisted factorization boundary.
- In the exceptional arm, D'_2, D'_3 follow analytically and D'_4, D'_5 have complete finite certificates. D'_6 retains a source-assisted primality boundary.
- Two-block tensor arguments prove the multicomponent zero arm at $h = 12, 15, 24, 36, 40$. These are theorem-level finite cases, not a universal generation result.

Exact finite screens also provide countermodels to claims based only on trace, norm, factor shape, conductor, or unselected Kummer data. Such a countermodel eliminates a proof route; it is not a counterexample to Conjecture 1.1 unless it lies on the literal Conway ancestry.

11 Conclusion

The finite excess problem has a precise candidate and a precise proof boundary. The 0/1/4 rule is equivalent to four universal selected-order assertions. In each arm the ambient field theory has been reduced to one marked Kummer coordinate, with the full primary valuation retained.

The reductions also explain why the conjecture resists coarse invariants. The zero arm needs the synchronized rather than componentwise phase; the ordinary arm needs one Artin symbol on the Conway-selected inert ray; the cubic and exceptional arms need the faithful finite reduction of a marked unit, including its principal-ray factor. Generic structure reconstructs these coordinates but does not force their nonvanishing. Until all four selected assertions are proved, both Conjecture 1.1 and absolute boundedness remain open.

References

- [1] Francesco Amoroso and Roberto Dvornicich. A lower bound for the height in abelian extensions. *Journal of Number Theory*, 80(2):260–272, 2000.
- [2] Leandro Cagliero, Allen Herman, and Fernando Szechtman. Artin–Schreier towers of finite fields. *Finite Fields and Their Applications*, 106:102606, 2025.
- [3] John H. Conway. *On Numbers and Games*. Academic Press, London, 1976.
- [4] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In André Platzer and Geoff Sutcliffe, editors, *Automated Deduction—CADE 28*, volume 12699 of *Lecture Notes in Computer Science*, pages 625–635, Cham, 2021. Springer.
- [5] Joseph DiMuro. On On_p , 2011.
- [6] Klaus Hoechsmann. Cyclotomic units over finite fields. *Rendiconti del Circolo Matematico di Palermo*, 44:5–20, 1995.
- [7] H. W. Lenstra, Jr. On the algebraic closure of two. *Indagationes Mathematicae (Proceedings)*, 80(5):389–396, 1977.
- [8] H. W. Lenstra, Jr. Nim multiplication. *Séminaire de Théorie des Nombres de Bordeaux*, 7, 1977–1978. Exposé 11, pp. 1–24.
- [9] Rudolf Lidl and Harald Niederreiter. *Finite Fields*, volume 20 of *Encyclopedia of Mathematics and its Applications*. Cambridge University Press, 2nd edition, 1997.
- [10] OEIS Foundation Inc. Entry A380496: Lenstra excess of the n -th odd prime. The On-Line Encyclopedia of Integer Sequences, 2026. Accessed 12 August 2026.
- [11] Roman Popovych. Multiplicative orders of elements in Conway’s towers of finite fields. *Algebra and Discrete Mathematics*, 25(1):137–146, 2018.
- [12] Michael Rosen. *Number Theory in Function Fields*, volume 210 of *Graduate Texts in Mathematics*. Springer, New York, 2002.
- [13] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, pages 367–381, New York, NY, USA, 2020. Association for Computing Machinery.
- [14] Tsuyoshi Uehara. On cyclotomic units connected with p -adic characters. *Journal of the Mathematical Society of Japan*, 37(1):65–77, 1985.